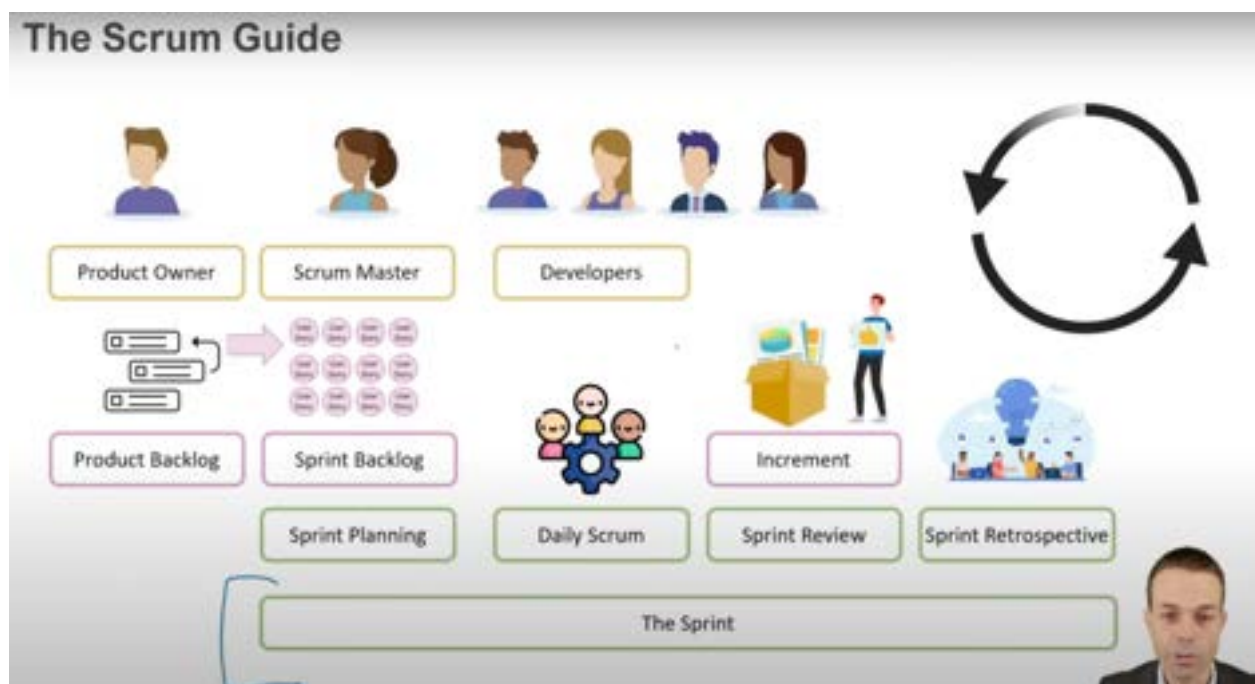
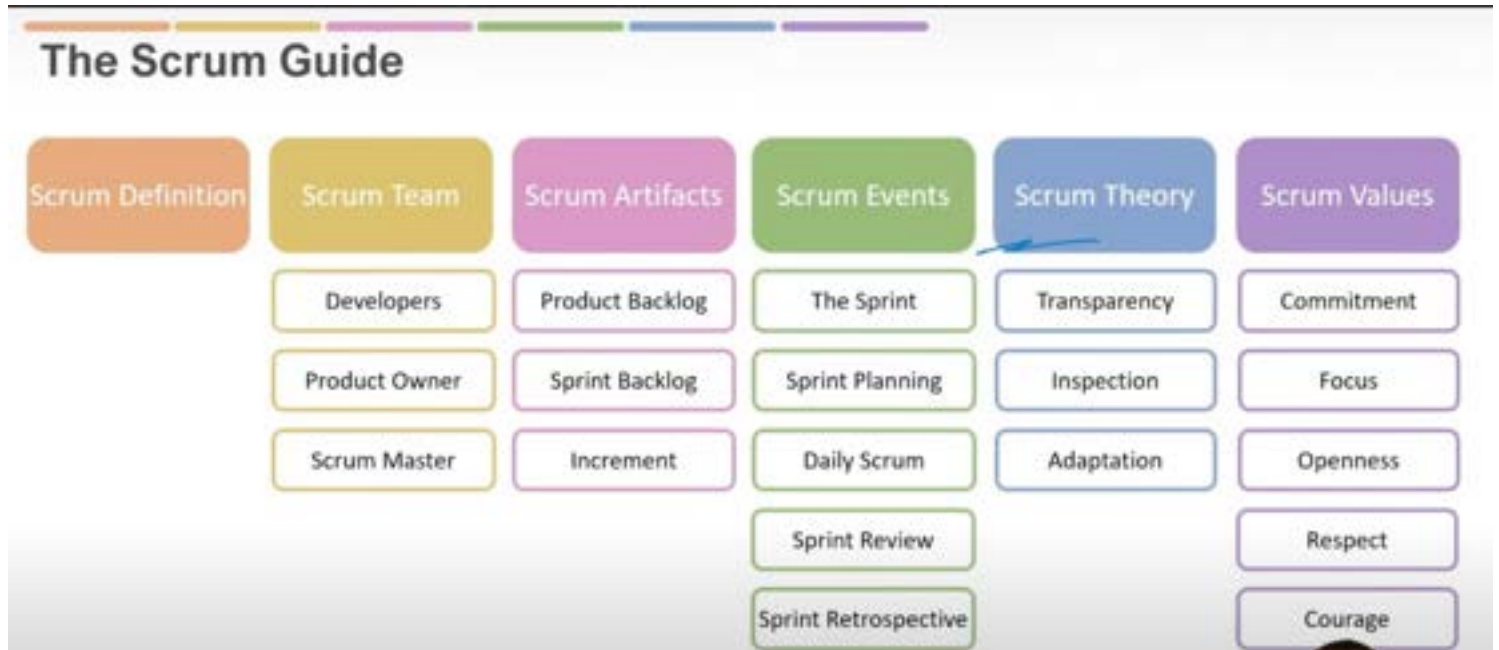


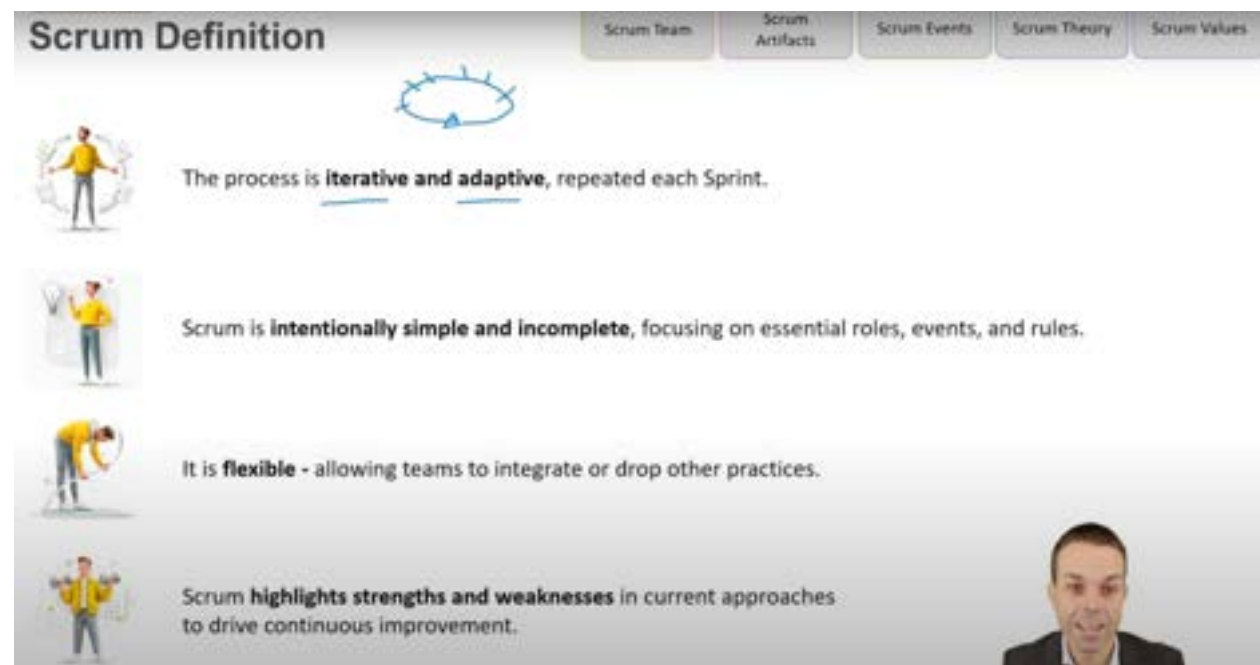
Table of Contents

Scrum Master Certification Guide	2
Scrum Definition	3
Scrum Team	4
Developers	6
Product Owner	6
Scrum Master	7
Scrum Artifacts	9
Product Backlog	10
Sprint Backlog	12
Increment	13
Scrum Events	15
The Sprint	16
Product Backlog Refinement	17
Sprint Planning	18
Daily Scrum	20
Sprint Review	21
Sprint Retrospective	23
Scrum Theory	24
Transparency	25
Inspection	26
Adaptation	26
Scrum Values	27

Scrum Master Certification Guide



Scrum Definition



Scrum Team



Scrum Team

Scrum
Definition

Scrum
Artifacts

Scrum Events

Scrum Theory

Scrum Values



Scrum Teams are:

- **Cross-functional** – they have all the skills needed to deliver value.
- **Self-managing** – they decide internally who does what, when, and how.

Optimal size is **10 or fewer people** to stay nimble and productive; larger teams should split while sharing:

- One Product Goal
- One Product Backlog
- One Product Owner



Scrum Team

Scrum
Definition

Scrum
Artifacts

Scrum Events

Scrum Theory

Scrum Values

The team is responsible for **all product-related activities**, from development to maintenance and experimentation.

Scrum Teams are **empowered by the organization** to manage their own work.

Working in Sprints, at a **sustainable pace** helps maintain focus and consistency.

The **whole team is accountable** for delivering a valuable **Increment** every



Developers

Scrum Team

Developers

Developers are Scrum Team members committed to delivering a **usable Increment** each Sprint.

Required skills vary depending on the **domain**, but Developers are always accountable for:

- Creating the **Sprint plan** (Sprint Backlog)
- **Ensuring quality** by following the Definition of Done
- Adjusting the plan daily to stay aligned with the **Sprint Goal**
- Holding each other accountable as professionals



Product Owner

Scrum Team

Product Owner

The **Product Owner** is accountable for **maximizing the product value** created by the Scrum Team.

This role varies by **organization, team, and individual**, but always includes **Product Backlog management**, such as:

- Defining and communicating the Product Goal
- Creating and clearly describing Backlog items
- Ordering Backlog items by priority
- Ensuring the Backlog is transparent and understood by all



The Product Owner may **delegate tasks**, but **remains accountable** for the outcome.



Scrum Team

Product Owner



The **organization must respect the Product Owner's decisions**, which are visible in:

- The **Product Backlog**
- The **Increment reviewed at the Sprint Review**

The Product Owner is **one person**, not a committee, and may represent many stakeholders.

Others can **suggest changes**, but only by **convincing the Product Owner**.

Scrum Master

Scrum Master



The **Scrum Master is accountable for establishing Scrum** as defined in the Scrum Guide.

They **promote understanding of Scrum theory and practices** across the Scrum Team and organization.

Responsible for the **Scrum Team's effectiveness** by helping improve their use of Scrum.



Scrum Team

Scrum
Definition

Scrum
Artifacts

Scrum Events

Scrum Theory

Scrum Values

Scrum Master

The Scrum Master serves the Scrum Team by:

- **Coaching** on self-management and cross-functionality.
- Helping the **team focus on high-value Increments** that meet the Definition of Done.
- **Removing impediments** to progress.
- Ensuring **Scrum events are effective**



Scrum Team

Scrum
Definition

Scrum
Artifacts

Scrum Events

Scrum Theory

Scrum Values

Scrum Master



The Scrum Master serves the Product Owner by:

- Supporting **effective Product Goal definition** and Backlog management.
- Helping the team understand **clear Product Backlog items**.
- Supporting **empirical product planning** in complex environments.
- **Facilitating stakeholder collaboration** as needed.

Scrum Team

Scrum
Definition

Scrum
Artifacts

Scrum Events

Scrum Theory

Scrum Values

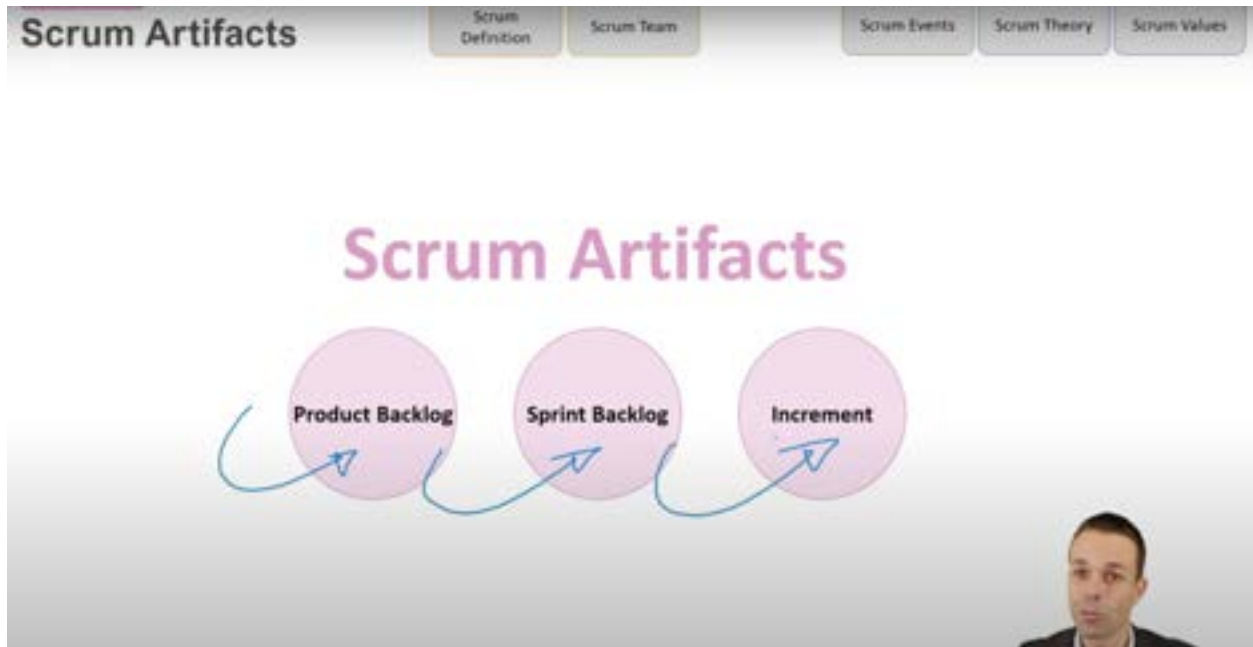
Scrum Master

The Scrum Master serves the organization by:

- **Leading, training, and coaching** on Scrum adoption.
- **Advising and planning** Scrum implementation.
- Helping others **embrace empirical approaches** for complex work.
- **Removing barriers** between stakeholders and Scrum Teams.



Scrum Artifacts



Scrum Artifacts

Scrum Definition Scrum Team Scrum Events Scrum Theory Scrum Values

Scrum **artifacts** represent **work** or **value** and are designed to ensure **transparency**.

Transparency ensures that **everyone shares the same understanding** for effective inspection and adaptation.

An illustration showing four stylized people (two men and two women) standing around a large magnifying glass. The magnifying glass is focused on a whiteboard that has various colorful sticky notes and diagrams on it. The people appear to be in a collaborative meeting, inspecting the work on the board.

Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

Each **artifact** includes a **commitment** to enhance transparency and focus:

1. Product Backlog



committed to the
Product Goal

2. Sprint Backlog



committed to the
Sprint Goal

3. Increment



committed to the
Definition of Done

Product Backlog

Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

1. Product Backlog

A **Product** is a vehicle for delivering value and should have:

- A **clear boundary**
- Known **stakeholders**
- Defined **users/customers**
- It can be a service, physical item, or something abstract



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

1. Product Backlog

The **Product Backlog** is an **emergent, ordered list** of everything needed to improve the product.

It is the **single source of work** for the Scrum Team.



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

1. Product Backlog

Commitment: Product Goal

- The **Product Goal** describes the **future state of the product** and acts as a target for planning.
- The Product Backlog evolves to deliver **what** will achieve the Product Goal.
- The team must **complete or abandon** one Product Goal before starting a new one.



Sprint Backlog

Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

2. Sprint Backlog

The **Sprint Backlog** is:

- The **Sprint Goal** (what are we delivering and **why** is it valuable?)
- Selected **Product Backlog items** for the Sprint (**what**)
- An **actionable plan** to deliver the Increment (**how**)



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

2. Sprint Backlog

It is created **by and for the Developers**.

Serves as a **real-time, visible plan** of the work for the Sprint.

It is **continuously updated** throughout the Sprint as more is learned.

Should contain **enough detail** to allow Developers to **inspect progress daily** during the **Daily Scrum**.



Scrum Artifacts

Scrum Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

2. Sprint Backlog

Commitment: Sprint Goal

- The **Sprint Goal** is the **single objective** for the Sprint.
- It is a **commitment by the Developers**.
- It **aligns** and **focuses** the Scrum Team on a shared purpose, avoiding scattered efforts.
- It provides **flexibility** in how the work is done to achieve the goal.
- If work changes, scope can be **renegotiated with the Product Owner**, but the **Sprint Goal remains unchanged**.



Increment

Scrum Artifacts

Scrum Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

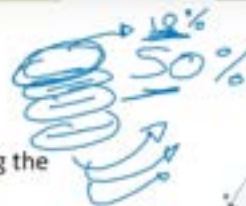
3. Increment

An **Increment** is a concrete step toward achieving the **Product Goal**.

Each Increment:

- Builds on all previous Increments
- Is thoroughly verified to ensure they all work together

To provide value, the Increment must be **usable**.



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

3. Increment

There can be **multiple Increments in a single Sprint**.

The **sum of all Increments** is presented at the **Sprint Review** to support **empiricism**.

The **Sprint Review** is **not a release gate** – value can be released any time.

Work is only part of an Increment if it **meets the Definition of Done**.



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

3. Increment

Commitment: Definition of Done

The **Definition of Done (DoD)** is a **formal description** of the required **quality standard** for an Increment.

When a Product Backlog item meets the DoD, **an Increment is created**.

It provides **transparency** by defining what “done” means for all stakeholders.



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

3. Increment

Commitment: Definition of Done

If a backlog item **does not meet the DoD**, it:

- Cannot be released
- **Cannot be presented** at the Sprint Review
- Is returned to the **Product Backlog**



Scrum Artifacts

Scrum
Definition

Scrum Team

Scrum Events

Scrum Theory

Scrum Values

3. Increment

Commitment: Definition of Done

If the organization has a standard DoD, **all Scrum Teams must follow it** at minimum.

If there is no organizational standard, the **Scrum Team defines its own DoD**.

When **multiple Scrum Teams** work on the same product, they must **agree on a shared DoD**.

Scrum Events

Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Scrum Events

The Sprint

Sprint Planning

Daily Scrum

Sprint Review

Sprint
Retrospective

Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

- Each Scrum event is a **formal opportunity to inspect and adapt** the artifacts. Skipping or altering events leads to missed chances for inspection and adaptation.
- Events are designed to **promote transparency** and continuous improvement.
- Scrum events help create regularity and **reduce the need for additional meetings**.
- Holding events at the same time and place helps reduce complexity.



Sprint Planning

Daily Scrum

Sprint Review

Sprint Retrospective

The Sprint



The Sprint

Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

The Sprint

- A Sprint is like a **short project**, offering clear timeframes and focus.
- The **Sprint is the container** for all other Scrum events.
- **Shorter Sprints** reduce complexity, risk, and increase learning opportunities.
- Forecasting tools (e.g., **burn-downs**, **burn-ups**, **cumulative flow**) can help - but do not replace **empiricism**. In complex environments, only past outcomes can guide future decisions.
- A Sprint can be **cancelled only by the Product Owner**, and only if the Sprint Goal becomes obsolete.



The sprint

1 - 4



Scrum Events

Scrum Definition

Scrum Team

Scrum Artifacts

Scrum Theory

Scrum Values

The Sprint

Sprints are the heartbeat of Scrum, where ideas are transformed into value.

They are **fixed-length events (max 1 month)** that start immediately after the previous Sprint ends.

During the Sprint:

- The **Sprint Goal** must remain stable
- Quality must not decrease
- The **Product Backlog** can be refined
- **Scope** can be clarified or renegotiated with the Product Owner as more is learned



The Sprint

Product Backlog Refinement

Scrum Events

Scrum Definition

Scrum Team

Scrum Artifacts

Scrum Theory

Scrum Values

Product Backlog Refinement

Product Backlog refinement is not an official event, but is an ongoing activity where:

- The team **break down** and **define** backlog items more precisely.
- They add **details** such as description, order, and size.
- **Developers** are responsible for **sizing** the items.
- The **Product Owner** supports by **clarifying trade-offs** and helping Developers understand priorities.



Product Backlog Refinement

Sprint Planning

Scrum Events

Scrum Definition

Scrum Team

Scrum Artifacts

Scrum Theory

Scrum Values

Sprint Planning

- **Timeboxed to 8 hours** for a one-month Sprint (shorter Sprints = shorter planning),
- Sprint Planning **starts the Sprint** by creating a **collaborative plan** for the work ahead.
- The **entire Scrum Team** participates, and others may be invited for expert advice.
- The **Product Owner** ensures the team is ready to discuss high-priority **Product Backlog** items and how they support the Product Goal.
- Items that can be **completed within one Sprint** are considered **"ready"** for Sprint Planning.



Sprint Planning

The Sprint



Scrum Events

Scrum Definition

Scrum Team

Scrum Artifacts

Scrum Theory

Scrum Values

Sprint Planning

Sprint Planning covers three main topics:

Why is this Sprint valuable?

What can be done this Sprint?

How will the chosen work get done?



Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Sprint Planning

Why is this Sprint
valuable?

What can be done
this Sprint?

How will the chosen
work get done?

- The **team** collaboratively defines the **Sprint Goal**, explaining the Sprint's value to stakeholders. The **Product Owner** gives guidance on how value could be increased.
- The Sprint Goal must be finalized during this event.

Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Sprint Planning

Why is this Sprint
valuable?

What can be done
this Sprint?

How will the chosen
work get done?

- **Developers** and the **Product Owner** select **Product Backlog** items to include.
- Items may be refined during planning to build understanding and confidence.
- Accurate forecasting depends on past performance, current capacity, and the Definition of Done.

Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Sprint Planning

Why is this Sprint
valuable?

What can be done
this Sprint?

How will the chosen
work get done?

- Developers plan how to turn selected Backlog items into a usable Increment.
- This often involves breaking items down into daily-sized tasks.
- Developers have full autonomy over how they do the work – no one tells them how.

Daily Scrum

Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Daily Scrum

The **Daily Scrum** is a **15-minute daily event** for Developers to:

- **Inspect progress** toward the Sprint Goal.
- **Adapt** the Sprint Backlog and update the plan for the next 24 hours.

It is held at the same time and place each working day to reduce complexity.



Sprint Planning

Daily Scrum



Scrum Events

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Daily Scrum

If the **Product Owner** or **Scrum Master** are working on Sprint Backlog items, they **participate as Developers**.

Developers **choose their own structure or techniques**, as long as:

- The event stays focused on **Sprint Goal progress**.
- It results in an **actionable plan for the next day**.



Scrum Events

Daily Scrum

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

Benefits of the Daily Scrum:



Developers can re-plan any time during the day – the Daily Scrum is not the only time for updates.

Sprint Review

Scrum Events

Sprint Review

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

The **purpose** of the Sprint Review is to:

- Inspect the Sprint's outcome –
- **Decide what to do next** based on progress and changes



Sprint Planning

Daily Scrum

Sprint Review



Scrum Events

Scrum
Definition

Scrum
Team

Scrum
Artifacts

Scrum
Theory

Scrum
Values

Sprint Review

The **Scrum Team** **presents** the results of their work to **key stakeholders**.

Together, the Scrum Team and stakeholders:

- Review what was accomplished
- Discuss progress toward the Product Goal
- Identify changes in the environment
- Collaborate on next steps



The **Product Backlog** may be updated to reflect new insights and opportunities.

Scrum Events

Scrum
Definition

Scrum
Team

Scrum
Artifacts

Scrum
Theory

Scrum
Values

Sprint Review



2w

This is a **working session**, not just a presentation – **active collaboration** is encouraged.

The Sprint Review is the **second-to-last event** in the Sprint.

It is **timeboxed to 4 hours** for a one-month Sprint (shorter if the Sprint is shorter).



8hr

15min

4hrs

Sprint Planning

Daily Scrum

Sprint Review



Sprint Retrospective

Scrum Events

Sprint Retrospective

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

The **Retrospective** helps us plan improvements in quality and effectiveness.

The Scrum Team **inspects** how the last Sprint went in terms of:

- Individuals and interactions
- Processes and tools
- Definition of Done



Sprint Planning

Daily Scrum

Sprint Review

Sprint Retrospective



Scrum Events

Sprint Retrospective

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Theory

Scrum Values

The team:

- Identifies **assumptions** that led them off track
- Explores their origins
- Reviews **what went well, what problems occurred, and how they were handled**
- The **most impactful improvements** are implemented quickly. These changes may be **added to the next Sprint Backlog**.



Scrum Events

Sprint Retrospective

The Sprint Retrospective is the **final event** in the Sprint.

Timeboxed to 3 hours for a one-month Sprint (shorter for shorter Sprints).



Scrum Theory

Scrum Theory

Scrum Theory

Transparency

Inspection

Adaptation

Scrum Theory

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Events

Scrum Values

Scrum is based on:

Empiricism (learning through experience and observation) and
Lean thinking (eliminating waste and focusing on what's essential).

It uses an **iterative, incremental approach** to improve predictability and manage risk.

Scrum Teams are cross-functional, with all the skills needed to complete the work—or the ability to acquire them.



Transparency

Scrum Theory

Scrum
Definition

Scrum Team

Scrum
Artifacts

Scrum Events

Scrum Values

1. Transparency

- ▶ **Work and processes must be visible** to both those doing the work and those receiving it.
- ▶ Scrum relies on the **transparency** of its three formal artifacts to support informed decision-making (Product Backlog, Sprint Backlog, Increment).
- ▶ **Low transparency** can result in poor decisions, reduced value, and increased risk.
- ▶ **Transparency enables effective inspection**; without it, inspection is **misleading and wasteful**.

Inspection

Scrum Theory

Scrum Definition

Scrum Team

Scrum Artifacts

Scrum Events

Scrum Values

2. Inspection

- **Scrum artifacts** and **progress toward goals** must be regularly and thoroughly inspected to identify issues or deviations.
- Scrum uses **five events** (the sprint, sprint planning, daily scrum, sprint review and sprint retrospective) to create a **regular cadence for inspection**.
- **Inspection supports adaptation** – without adaptation, inspection has no value.
- **Scrum events are intended to drive change** and continuous improvement.



Adaptation

Scrum Theory

Scrum Definition

Scrum Team

Scrum Artifacts

Scrum Events

Scrum Values

3. Adaptation



If a **process** or **product** falls outside acceptable limits, it must be **adjusted quickly** to avoid further issues.

Timely adaptation is essential to minimize deviation.

Adaptation is harder when teams are not **empowered** or **self-managing**.

Scrum Teams are expected to **adapt immediately** based on what they learn through inspection.

Scrum Values

Scrum Values

Scrum DefinitionScrum TeamScrum ArtifactsScrum EventsScrum Theory

Scrum success depends on living the five core values:

Commitment	Focus	Openness	Respect	Courage
				
to goals and team support	on Sprint work and progress	about work and challenges	for each team member's capabilities and independence	to face difficult problems and do what's right



Scrum Values

Scrum DefinitionScrum TeamScrum ArtifactsScrum EventsScrum Theory

These values guide the team's **work, behavior, and decisions**.

Scrum practices should **reinforce** – not undermine – **these values**.

As teams embody the values, the **Scrum pillars of transparency, inspection, and adaptation** come to life and build trust.

